

Effective Coefficients of the Wave Equation under PML Transformation

Junkyo Seo

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1 Strong and Weak Forms of Boundary Value Problems

1.1 Examples of Strong and Weak Forms

Consider the following one-dimensional boundary value problem defined on the open interval $(0, 1)$:

$$\begin{cases} \frac{d^2 u}{dx^2} + f = 0, & x \in (0, 1), \\ u = 0, & x = 0, \\ u = 0, & x = 1, \end{cases} \quad (1)$$

where $u(x)$ denotes the unknown scalar field and $f(x)$ is a given source term.

Equation (1) represents the *strong form* of the boundary value problem. In this formulation, the governing differential equation and boundary conditions are required to be satisfied pointwise in the classical sense. Consequently, the solution u must possess sufficient smoothness such that the second-order derivative exists almost everywhere in the domain, typically requiring $u \in C^2(0, 1)$.

Here, an important point to note is that the zero on the right-hand side is not merely a numerical constant but rather a *zero function* whose inner product with any other function always yields zero. Using this property, if we take the inner product of both sides of Equation (1) with an arbitrary function v , the resulting expression can be written as follows.

$$(v, \frac{d^2 u}{dx^2} + f) = \int_0^1 v (\frac{d^2 u}{dx^2} + f) dx = 0 \quad (2)$$

Let us assume that $v = 0$ on the boundary of the domain. By applying integration by parts, the expression can be rearranged into the following form.

$$\begin{aligned} 0 &= \int_0^1 v \left(\frac{d^2 u}{dx^2} + f \right) dx \\ &= \left[v \frac{du}{dx} \right]_0^1 - \int_0^1 \frac{dv}{dx} \frac{du}{dx} dx + \int_0^1 v f dx \\ &= - \int_0^1 \frac{dv}{dx} \frac{du}{dx} dx + \int_0^1 v f dx. \end{aligned} \quad (3)$$

In the above, v is called *test function* and u is called *trial function*. Using this result, the *weak form* of the same problem can be defined as follows.

Given f , find $u \in \mathcal{U}$ such that $\forall v \in \mathcal{U}$,

$$\int_0^1 \frac{dv}{dx} \frac{du}{dx} dx = \int_0^1 v f dx \quad (4)$$

Here, $\mathcal{U}$ is a function space such that

$$\mathcal{U} = \{u \in H^1(0, 1) ; u(0) = u(1) = 0\}. \quad (5)$$

$H^k(0, 1)$ is a Sobolev space, such that

$$H^k(0, 1) = \left\{ u ; u \in L_2(0, 1), \frac{du}{dx} \in L_2(0, 1), \dots, \frac{d^k u}{dx^k} \in L_2(0, 1) \right\}. \quad (6)$$

where

$$L_2(0, 1) = \left\{ u ; \int_0^1 u^2 dx < \infty \right\}. \quad (7)$$

The reason we define the function spaces using Sobolev spaces in the following manner is to ensure square integrability while preserving the boundary conditions, thereby guaranteeing the numerical stability of the solution to the weak form. Looking at the left-hand side of Equation (4), It can be observed that it is expressed as the integral of the product of the first derivatives of the two functions u and v with respect to x . Here, the advantage of employing Sobolev spaces becomes evident: this framework allows us to readily ensure that the corresponding quantity does not diverge. Since both u and v belong to $H^1(0, 1)$, they satisfy the following property.

$$A = \int_0^1 \left(\frac{du}{dx} \right)^2 dx < \infty \quad (8)$$

$$B = \int_0^1 \left(\frac{dv}{dx} \right)^2 dx < \infty \quad (9)$$

By applying the Cauchy–Schwarz inequality, the term of interest can be bounded and expressed in the following inequality form.

$$\left| \int_0^1 \frac{dv}{dx} \frac{du}{dx} dx \right| \leq \underbrace{\sqrt{\int_0^1 \left(\frac{dv}{dx} \right)^2 dx}}_{\sqrt{B} \text{ (finite)}} \underbrace{\sqrt{\int_0^1 \left(\frac{du}{dx} \right)^2 dx}}_{\sqrt{A} \text{ (finite)}}. \quad (10)$$

By the inequality, the left-hand side (the integral value of interest) is bounded above by the right-hand side, and is therefore guaranteed to be finite.

1.2 Conceptual Differences Between Strong and Weak Forms

The strong and weak formulations of boundary value problems differ not only in their mathematical representations but also in their underlying conceptual and analytical foundations. As demonstrated in the preceding subsection, the strong form expresses the governing differential equation and boundary conditions in a pointwise sense, requiring the solution to possess sufficient classical smoothness. In contrast, the weak form reformulates the same problem within an integral framework, thereby relaxing the regularity requirements imposed on the solution.

A fundamental distinction lies in the manner in which derivatives are treated. In the strong form, the differential operator acts directly on the solution, and the existence of classical derivatives is implicitly assumed. For the one-dimensional problem considered in Equation (1), this necessitates that the solution admits a second-order derivative almost everywhere in the domain. Such requirements can be overly restrictive, particularly in problems involving non-smooth domains, discontinuous coefficients, or singular source terms.

The weak form addresses this limitation by transferring derivatives from the trial function to the test function through integration by parts. As a result, the highest-order derivatives appearing in the formulation are of lower order than those in the strong form. In the present example, the second-order derivative in the governing equation is replaced by first-order derivatives in the bilinear form, as shown

in Equation (4). This reduction in derivative order enables the solution to be sought in a Sobolev space rather than in a space of twice continuously differentiable functions.

Another key conceptual difference concerns the interpretation of boundary conditions. In the strong form, boundary conditions are imposed explicitly and must be satisfied pointwise. In the weak formulation, essential (Dirichlet) boundary conditions are incorporated into the definition of the function space, while the governing equation itself is enforced in an averaged sense over the domain. This distinction is reflected in the choice of the space $\mathcal{U} = H_0^1(0,1)$, which ensures that the boundary conditions are satisfied inherently by all admissible trial and test functions.

Finally, the weak form plays a central role in numerical approximation methods. Its integral structure and reliance on Sobolev spaces form the theoretical foundation of the finite element method, where solutions are approximated within finite-dimensional subspaces. In contrast, the strong form, while conceptually straightforward, is generally unsuitable as a direct starting point for numerical discretization.

In summary, the strong form represents the classical, pointwise expression of a boundary value problem, whereas the weak form provides a generalized, variational representation that accommodates weaker solution regularity and facilitates both theoretical analysis and numerical computation.

2 Application of the PML Transformation to the Wave Equation: Strong and Weak Forms

2.1 Derivation of the General Form of the Wave Equation: The SH-Wave Case

In general, the wave equation for a source-free scalar wave can be written as follows.

$$\frac{1}{b} \frac{\partial^2 u}{\partial t^2} = \nabla \cdot (a \nabla u) \quad (11)$$

In this equation, the coefficients a and b represent material properties associated with wave propagation. In general, a corresponds to the stiffness of the medium, while b represents its inertia. The reason is that wave equations are generally derived from a combination of *Newton's second law* and *Hooke's law*. As a representative example, let us consider the case of a shear horizontal wave which is the simplest case of elastic wave.

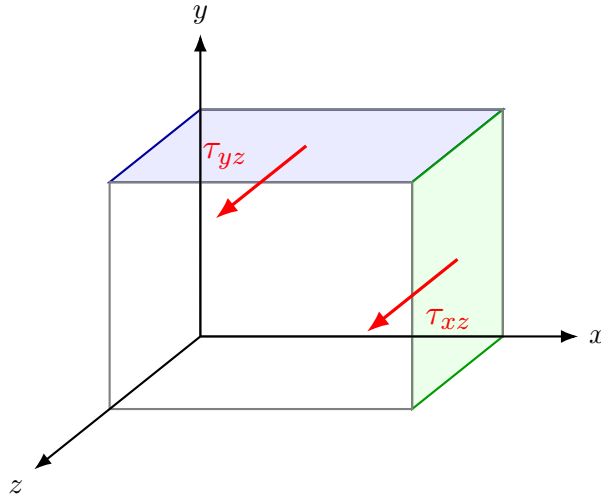


Figure 1: Infinitesimal volume element dV and the shear stress τ_{*z}

From this point onward, as illustrated in the figure above, let us consider an infinitesimal volume element with dimensions dx in the x -direction, dy in the y -direction, and unit length in the z -direction. Since we focus on a shear horizontal (SH) wave, we can assume that the displacement exists only in the z -direction.

First, let us evaluate the shear forces acting on this infinitesimal volume element. On the left face at x and the right face at $x + dx$, a shear stress τ_{xz} acting in the z -direction is applied. Since the corresponding area at each location is dy , the shear force contribution in the x -direction can be evaluated as follows.

$$dF_x = (\tau_{xz}(x + dx, y) - \tau_{xz}(x, y))dy \quad (12)$$

Assuming that dx is sufficiently small, we may approximate $\tau_{xz}(x + dx, y)$ using a first-order Taylor expansion. Based on this approximation, the expression can be rearranged and is ultimately written as follows.

$$dF_x = \frac{\partial \tau_{xz}}{\partial x} dx dy \quad (13)$$

The stress contribution in the y -direction can be evaluated in the same manner, yielding an analogous result. Therefore, the total stress acting in the z -direction can be expressed as follows.

$$F_z = \left(\frac{\partial \tau_{xz}}{\partial x} + \frac{\partial \tau_{yz}}{\partial y} \right) dx dy = (\nabla \cdot \boldsymbol{\tau}) dx dy \quad (14)$$

Here, $\boldsymbol{\tau}$ denotes the two-dimensional shear stress vector associated with the z -direction. Let the density of the infinitesimal region be denoted by ρ , and let the displacement in the z -direction be $u(x, y, t)$. By applying Newton's second law and rearranging the terms, the resulting expression can be written as follows.

$$\nabla \cdot \boldsymbol{\tau} dx dy = \rho dx dy \frac{\partial^2 u}{\partial t^2} \quad (15)$$

By dividing both sides by the unit volume element dx, dy , the final expression is obtained as follows.

$$\nabla \cdot \boldsymbol{\tau} = \rho \frac{\partial^2 u}{\partial t^2} \quad (16)$$

From this point onward, we express $\boldsymbol{\tau}$ in terms of the displacement u by employing the stress-strain relationship, which is a form of Hooke's law applicable to the material.

Strain ϵ is tensor-valued quantity that represents how an infinitesimal element deforms due to stress. More rigorously, it is the spatial rate of displacement $r = (r_x, r_y, r_z)$, in this case $r = (0, 0, u)$. In linear elasticity, strain is the symmetric part of the displacement gradient, i.e.,

$$\epsilon = \begin{pmatrix} \epsilon_{xx} & \epsilon_{xy} & \epsilon_{xz} \\ \epsilon_{yz} & \epsilon_{yy} & \epsilon_{yz} \\ \epsilon_{zx} & \epsilon_{zy} & \epsilon_{zz} \end{pmatrix} = \text{grad}_{sym} \mathbf{r} = \frac{1}{2} (\text{grad } \mathbf{r} + (\text{grad } \mathbf{r})^T) \quad (17)$$

Note that rigid translations and rotations do not induce stress. Therefore, they are removed from the strain by taking the gradient of the displacement and subsequently extracting its symmetric part. Using the result of the above expression, the strains ϵ_{xz} and ϵ_{yz} can be computed, yielding the following results.

$$\begin{aligned} \epsilon_{xz} &= \frac{1}{2} \frac{\partial u}{\partial x} \\ \epsilon_{yz} &= \frac{1}{2} \frac{\partial u}{\partial y} \end{aligned} \quad (18)$$

By employing the constitutive relationship between shear stress and strain, each expression can be rearranged and written as follows.

$$\boldsymbol{\tau} = \begin{pmatrix} \tau_{xz} \\ \tau_{yz} \end{pmatrix} = \begin{pmatrix} 2\mu\epsilon_{xz} \\ 2\mu\epsilon_{yz} \end{pmatrix} = \mu \begin{pmatrix} \frac{\partial u}{\partial x} \\ \frac{\partial u}{\partial y} \end{pmatrix} = \mu \nabla u \quad (19)$$

Substituting $\boldsymbol{\tau}$ into the Equation (16), the desired form of the wave equation can be obtained in the end.

$$\nabla \cdot (\mu \nabla u) = \rho \frac{\partial^2 u}{\partial t^2} \quad (20)$$

In this case, a corresponds to the shear modulus μ , which is a stiffness-related term, while b corresponds to the inverse of the density $\left(\frac{1}{\rho}\right)$, i.e., the specific volume, which represents an inertia-related term.

2.2 Determination of Effective Coefficients under the PML Transformation

In this section, we derive the governing equation obtained by applying the PML transformation to the SH-wave equation derived earlier, and compute the corresponding effective coefficients appearing in that equation. Furthermore, we analyze the validity of the resulting formulations by applying the PML transformation separately to the *strong form* and the *weak form* of the governing equation.

For computational convenience, let us further assume that the material properties ρ and μ are both functions of x only.

2.2.1 Strong form

First, let us examine how the governing equation is modified when the PML transformation is applied to its *strong form*.

The fundamental principle of the PML is to exploit the fact that the solution is an analytic function of x , and to attenuate waves by analytically continuing the coordinate axis from the real domain into the complex domain. When this formulation is subsequently mapped back to the real coordinate system, the coefficients of the equation are modified into material parameters with complex-valued properties. Since the coordinate stretching in the PML is defined in the frequency domain, we first apply the Fourier transform to express the equation in the frequency domain.

$$\hat{u}(x, y, w) = \int_{-\infty}^{\infty} u(x, y, t) e^{-i\omega t} dt \quad (21)$$

By the properties of the Fourier transform, differential operators with respect to t are transformed as follows:

$$\begin{aligned} \frac{\partial u}{\partial t} &\longrightarrow -i\omega \hat{u}, \\ \frac{\partial^2 u}{\partial t^2} &\longrightarrow (-i\omega)^2 \hat{u} = -\omega^2 \hat{u}. \end{aligned} \quad (22)$$

Applying this property to the original governing equation, its form is modified as follows.

$$\nabla \cdot (\mu \nabla \hat{u}) + \rho \omega^2 \hat{u} = 0 \quad (23)$$

Let's denote the complex x contour by $\tilde{x}$. Accordingly, the x -coordinate appearing in the divergence term and the gradient operator on the left-hand side is replaced by $\tilde{x}$. Introducing the stretch factor $s_x = 1 + i\frac{\sigma_x}{\omega}$ which is used to map the coordinate x and $\tilde{x}$, each term can be expressed as follows.

$$\begin{aligned} \tilde{\nabla} \cdot (\mu \tilde{\nabla} \hat{u}) &= \frac{\partial}{\partial \tilde{x}} \left(\mu \frac{\partial \hat{u}}{\partial \tilde{x}} \right) + \frac{\partial}{\partial y} \left(\mu \frac{\partial \hat{u}}{\partial y} \right) \\ &= \frac{1}{s_x} \frac{\partial}{\partial x} \left(\frac{\mu}{s_x} \frac{\partial \hat{u}}{\partial x} \right) + \frac{\partial}{\partial y} \left(\mu \frac{\partial \hat{u}}{\partial y} \right) \end{aligned} \quad (24)$$

Substituting this result into the Equation (23) and rearranging the terms, we obtain,

$$\frac{1}{s_x} \frac{\partial}{\partial x} \left(\frac{\mu}{s_x} \frac{\partial \hat{u}}{\partial x} \right) + \frac{\partial}{\partial y} \left(\mu \frac{\partial \hat{u}}{\partial y} \right) + \rho \omega^2 \hat{u} = 0 \quad (25)$$

Multiplying both sides of the equation by s_x and rearranging the terms, we obtain

$$\frac{\partial}{\partial x} \left(\frac{\mu}{s_x} \frac{\partial \hat{u}}{\partial x} \right) + \frac{\partial}{\partial y} \left(\mu s_x \frac{\partial \hat{u}}{\partial y} \right) + \rho s_x \omega^2 \hat{u} = 0 \quad (26)$$

To simplify the notation, let us define the diagonal matrix

$$\mathbf{S} = \text{diag}(s_x, 1).$$

Rearranging the equation so that it takes the same form as the Equation (23), it can be written as follows.

$$\nabla \cdot (\det(\mathbf{S})\mathbf{S}^{-1}\mu\mathbf{S}^{-1}\nabla\hat{u}) + \det(\mathbf{S})\rho\omega^2\hat{u} = 0 \quad (27)$$

Consequently, the effective material properties in the PML-applied effective domain, namely the effective coefficients μ_{eff} and ρ_{eff} , are defined as follows.

$$\mu_{\text{eff}} = \det(\mathbf{S})\mathbf{S}^{-1}\mu\mathbf{S}^{-1} = \mu \begin{pmatrix} \frac{1}{s_x} & \\ & s_x \end{pmatrix}, \quad \rho_{\text{eff}} = \det(\mathbf{S})\rho = \rho s_x \quad (28)$$

Up to this point, we have expressed the equations at the level of partial differential operators in order to clearly illustrate how each operator is transformed, and how the governing equation is subsequently reformulated based on these transformations. However, a more general approach to this kind of problems is to interpret the PML transformation as a form of coordinate transformation. If the diagonal matrix $\mathbf{S}$ defined earlier is regarded as a coordinate transformation matrix, then it follows that the gradient and divergence operators can be expressed as follows.

$$\tilde{\nabla} = \mathbf{S}^{-1}\nabla, \quad \tilde{\nabla} \cdot \mathbf{F} = \frac{1}{J}\nabla \cdot (J\mathbf{S}^{-1}\mathbf{F}) \quad (29)$$

Here, J denotes the Jacobian of the transformation and is defined as the determinant of the transformation matrix $\mathbf{S}$. In the problem under consideration, this Jacobian reduces to $J = s_x$. Substituting these properties into the original governing equation and rearranging the terms, we obtain

$$\frac{1}{\det\mathbf{S}}\nabla \cdot (\det(\mathbf{S})\mathbf{S}^{-1}\mu\mathbf{S}^{-1}\nabla\hat{u}) + \rho\omega^2\hat{u} = 0 \quad (30)$$

Multiplying both sides of the equation by $\det(\mathbf{S})$, one can verify that the same result is obtained.

2.2.2 Weak form

In addition to the strong form derivation, the effective material properties can also be derived by formulating the problem in the *weak form*. This approach is particularly essential for numerical implementations such as the Finite Element Method (FEM).

We begin with the governing equation in the frequency domain, as defined in Equation (23):

$$\nabla \cdot (\mu\nabla\hat{u}) + \rho\omega^2\hat{u} = 0 \quad (31)$$

To obtain the weak form, we multiply the equation by an arbitrary test function and integrate over the domain :

$$\int_{\Omega} v [\nabla \cdot (\mu\nabla\hat{u}) + \rho\omega^2\hat{u}] d\Omega = 0 \quad (32)$$

Applying Green's first identity (integration by parts) to the first term yields:

$$-\int_{\Omega} \nabla v \cdot (\mu\nabla\hat{u}) d\Omega + \oint_{\Gamma} v\mu\frac{\partial\hat{u}}{\partial n} d\Gamma + \int_{\Omega} v\rho\omega^2\hat{u} d\Omega = 0 \quad (33)$$

Assuming that the boundary integral vanishes due to the homogeneous boundary conditions imposed on the test function or the flux, the weak form in the physical domain is simplified to:

$$\int_{\Omega} \nabla v \cdot (\mu\nabla\hat{u}) d\Omega - \int_{\Omega} v\rho\omega^2\hat{u} d\Omega = 0 \quad (34)$$

To apply the PML, we extend the coordinate system into the complex domain using the coordinate stretching defined previously. Consequently, the integration domain changes from the real domain to the complex domain, and the differential operator is replaced by the complex operator. The weak form in the PML domain is expressed as:

$$\int_{\tilde{\Omega}} \tilde{\nabla} v \cdot (\mu\tilde{\nabla}\hat{u}) d\tilde{\Omega} - \int_{\tilde{\Omega}} v\rho\omega^2\hat{u} d\tilde{\Omega} = 0 \quad (35)$$

To solve this equation numerically, we map the integral back to the real coordinate system . This transformation requires the modification of the differential operator and the differential volume element using the stretch factor matrix . The relations are given by:

$$\tilde{\nabla} = \mathbf{S}^{-1}\nabla, \quad d\tilde{\Omega} = \det(\mathbf{S})d\Omega \quad (36)$$

Substituting these relations into the integral equation, the stiffness term (the first term) is transformed as follows:

$$\int_{\tilde{\Omega}} (\tilde{\nabla} v)^T \mu (\tilde{\nabla} \hat{u}) d\tilde{\Omega} = \int_{\Omega} (\mathbf{S}^{-1}\nabla v)^T \mu (\mathbf{S}^{-1}\nabla \hat{u}) \det(\mathbf{S}) d\Omega = \int_{\Omega} (\nabla v)^T [\det(\mathbf{S})\mathbf{S}^{-1}\mu\mathbf{S}^{-1}] (\nabla \hat{u}) d\Omega \quad (37)$$

Similarly, the mass term (the second term) is transformed as:

$$\int_{\tilde{\Omega}} v \rho \omega^2 \hat{u} d\tilde{\Omega} = \int_{\Omega} v \rho \omega^2 \hat{u} \det(\mathbf{S}) d\Omega = \int_{\Omega} v [\det(\mathbf{S})\rho] \omega^2 \hat{u} d\Omega \quad (38)$$

By comparing the transformed terms with the standard weak form in Equation (34), we can identify the effective material properties. The effective density is given by:

$$\rho_{\text{eff}} = \det(\mathbf{S})\rho = s_x \rho \quad (39)$$

The effective stiffness, originally a scalar , becomes an anisotropic tensor in the PML region due to the coordinate stretching:

$$\mu_{\text{eff}} = \det(\mathbf{S})\mathbf{S}^{-1}\mu\mathbf{S}^{-1} = s_x \begin{bmatrix} \frac{1}{s_x} & \\ & 1 \end{bmatrix} \mu \begin{bmatrix} \frac{1}{s_x} & \\ & 1 \end{bmatrix} = \mu s_x \begin{bmatrix} \frac{1}{s_x^2} & \\ & 1 \end{bmatrix} = \mu \begin{bmatrix} \frac{1}{s_x} & \\ & s_x \end{bmatrix} \quad (40)$$

In conclusion, the application of PML each strong form and weak form results in an anisotropic effective stiffness tensor and a modified scalar density.